

Credit Name: CSE2920 - CSE Project C

Assignment Name: Tug Of War

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to create a code that will receive 10 inputs from either the red or green button (whichever button is pressed 10 times fastest is the winner), flash both red and green LEDs, and then flash the winner's LED 5 times.

CODING:

1.

```
import com.phidget22.DigitalInput;

public class TugOfWar {
    public static void main(String[] args) throws Exception{

        //Create | Create objects for your buttons and LEDs.
        DigitalInput redButton = new DigitalInput();
        DigitalOutput redLED = new DigitalOutput();
        DigitalInput greenButton = new DigitalInput();
        DigitalOutput greenLED = new DigitalOutput();

        //Address | Address your four objects which lets you
        redButton.setHubPort(0);
        redButton.setIsHubPortDevice(true);
        redLED.setHubPort(1);
        redLED.setIsHubPortDevice(true);
        greenButton.setHubPort(5);
        greenButton.setIsHubPortDevice(true);
        greenLED.setHubPort(4);
        greenLED.setIsHubPortDevice(true);

        //Open | Connect your program to your physical device
        redButton.open(1000);
        redLED.open(1000);
        greenButton.open(1000);
        greenLED.open(1000);
    }
}
```

Phidgets library is imported

greenButton, greenLED, redButton, and redLED objects were created to interact with the physical phidgets box.

The HubPort number was set based on the appropriate object.

IsHubPortDevice set to true for all objects (confirming to the program that the objects are part of phidgets box).

Opened all of the objects, establishing a connection to the physical Phidgets box.

2.

```
int redCount = 0, greenCount = 0;

while (redCount < 10 && greenCount < 10) {
    if (redButton.getState()) {
        redCount++;
        Thread.sleep(150);
    }
    if (greenButton.getState()) {
        greenCount++;
        Thread.sleep(150);
    }
}
```

Declared the variables redCount and greenCount, which will store the number of button presses depending on which colour button is pressed.

While redCount and greenCount are less than 10, each button press (red or green) will increase the redCount or greenCount variable by 1. Sleep time of 150ms so that button presses are accurate.

3.

```
if (redCount >= 10 || greenCount >= 10) {
    redLED.setState(true);
    greenLED.setState(true);
    Thread.sleep(400);
    redLED.setState(false);
    greenLED.setState(false);
    Thread.sleep(400);
}
```

If either redCount or greenCount reaches or exceeds 10, then both LEDs will flash ONCE.

Please note that when creating the video for Tug Of War, I had an error. Instead of the LEDs flashing once, I coded it to flash twice. I fixed this error and now the LEDs flash once.

4.

```
if (redCount >= 10) {
    for (int i = 0; i < 5; i++) {
        redLED.setState(true);
        Thread.sleep(300);
        redLED.setState(false);
        Thread.sleep(300);
    }
}
if (greenCount >= 10) {
    for (int i = 0; i < 5; i++) {
        greenLED.setState(true);
        Thread.sleep(300);
        greenLED.setState(false);
        Thread.sleep(300);
    }
}
```

The program differentiates which button was pressed 10 times first. If it was the red side that won, the red LED flashes on and off 5 times using a for loop. If it was the green side that won, the green LED flashes on and off 5 times using a for loop.

End Of Program!